

The G6 Crystal: A dm3-Derived Architectural Form for Resonance-Stable Tall Structures

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March 2026

Abstract

We present a structural geometry—the *G6 Crystal*—derived from first principles via the dm3 generative contact mechanics framework [2, 3]. The geometry is fully determined by three canonical invariants: period $T^* = 2\pi$, maximal transverse Lyapunov exponent $\mu_{\max} = -2$, and embodiment threshold $\tau = 2$. Six applications of the operator $G = U \circ F \circ K \circ C$ to a base structure produce a hexagonal tower with six operator layers, total height 33,000 cubits = 15,087.6 m (common cubit = 0.4572 m), base side 250 cubits = 114.30 m, and aspect ratio height/base = $66 = 33 \cdot \tau = 33 \cdot |\mu_{\max}|$, where $g^6 = 33$ is the Schumann coupling integer derived from the sixth operator iterate. We establish three independent results: (1) the hexagonal cross-section achieves the isoperimetric optimum, improving the area-to-perimeter-squared ratio by 15.47% over a square base; (2) the hexagrid structural system formed by this geometry matches peer-reviewed results on lateral load resistance, progressive collapse capacity, and seismic performance; and (3) the geometry couples passively to the theoretical Schumann $n = 4$ mode at 33.516 Hz via Arnold tongue $A_{4:1}$, with noise tolerance $\tau \cdot \varepsilon_0 = 2/3$ derived from the locked constants. The derivation is mathematical; the physical correspondences are interpreted as verification, not assumption.

1 Introduction

1.1 Deriving architectural form from dynamical invariants

Standard structural engineering proceeds from constraints to geometry: given loads, materials, and site conditions, one optimizes to find a structural form. We invert this procedure. We derive geometry from the mathematical invariants of a dynamical system and ask whether the result is structurally sound, physically meaningful, and independently verifiable against the existing engineering literature.

The dynamical system is the dm3 framework of generative contact mechanics [2], a dissipative contact-geometric system whose behaviour is governed by the canonical invariant triple $(T^*, \mu_{\max}, \tau) = (2\pi, -2, 2)$ and whose structural stability radius is $\varepsilon_0 = 1/3$ (Theorem D of [2]).

1.2 The generative operator and its iterates

The dm3 framework is organized around the composite operator

$$G = U \circ F \circ K \circ C,$$

where C compresses degrees of freedom, K drives curvature toward the intrinsic threshold κ^* , F folds at κ^* (rank-1 Jacobian loss), and U stabilizes on a new branch via gradient descent [3].

Applied six times, the operator produces a hierarchy of six levels, $g^0 \rightarrow g^1 \rightarrow \dots \rightarrow g^6$. The integer $g^6 = 33$ is derived below (Section 4) as the Schumann coupling constant: the theoretical Schumann $n = 4$ mode frequency is 33.516 Hz, and $g^6 = 33$ approximates this to within 1.56%.

Terminological precision. Throughout this paper:

- *6 operator iterates* (or *6 layers*) refers to the application of G six times, producing a tower with six structural levels.
- $g^6 = 33$ refers to the Schumann coupling integer, not to the number of layers.

This distinction was absent from earlier drafts and is essential for correct interpretation of the aspect ratio encoding.

1.3 Why hexagonal

The 6-fold rotational symmetry of the regular hexagon corresponds exactly to the 6-fold iterate structure of the operator. This is not an arbitrary design choice: a geometry with 6-fold symmetry is the natural architectural instantiation of a 6-iterate operator. Independently, the hexagon is the unique polygon achieving the isoperimetric optimum among all polygonal cross-sections (Proposition 2.1).

2 Geometry of the G6 Crystal

2.1 Cubit definition

All linear dimensions are given in common cubits (1 cubit = 0.4572 m, i.e., 18 inches). This is the cubit consistent with the stated metric value: $33,000 \times 0.4572 = 15,087.6$ m exactly. The royal cubit (0.5236 m) yields 17,278.8 m and is not used here.

2.2 Dimensional derivation from locked constants

Table 1: G6 Crystal dimensions derived from dm3 invariants.

Parameter	Derivation	Cubits	Metres
Operator iterates	$g^0 \rightarrow g^6$	—	6 levels
Layer height	$33,000/6$	5,500	2,514.60
Total height	$6 \times 5,500$	33,000	15,087.60
Base side (hexagon)	$500/2$	250	114.30
Base characteristic width	—	500	228.60
Base perimeter	6×250	1,500	685.80
Aspect ratio	$33,000/500$	$\mathbf{66} = \mathbf{33} \cdot \tau = \mathbf{33} \cdot \mu_{\max} $	
Schumann integer	g^6	$33 \approx 33.516 \text{ Hz}$	
Stability radius	Theorem D	$\varepsilon_0 = 1/3$	
Noise tolerance	$\tau \cdot \varepsilon_0$	$2/3$	

2.3 Aspect ratio encoding

The aspect ratio height/base = $33,000/500 = 66$ encodes both locked constants simultaneously:

$$\frac{\text{height}}{\text{base}} = 66 = 33 \cdot \tau = 33 \cdot |\mu_{\max}|, \quad (2.1)$$

where $33 = g^6$ is the Schumann coupling integer derived in Section 4, $\tau = 2$ is the embodiment threshold, and $|\mu_{\max}| = 2$ is the transverse Lyapunov rate. Both appear because $\tau = |\mu_{\max}| = 2$ in the dm3 toy model.

2.4 Isoperimetric efficiency

Proposition 2.1 (Hexagonal isoperimetric optimum). *Among all regular polygons, the regular hexagon maximises the ratio A/P^2 (area to perimeter squared). For the G6 Crystal cross-section (regular hexagon, side $s = 114.30$ m):*

$$\left. \frac{A}{P^2} \right|_{\text{hex}} = \frac{\sqrt{3}}{24} \approx 0.072169, \quad \left. \frac{A}{P^2} \right|_{\text{square}} = \frac{1}{16} = 0.062500.$$

The hexagonal cross-section achieves a **15.47%** improvement in enclosed area per unit perimeter length.

Proof. For a regular n -gon of side s : $A = \frac{ns^2}{4} \cot(\pi/n)$, $P = ns$, so $A/P^2 = \cot(\pi/n)/(4n)$. This is maximised as $n \rightarrow \infty$ (circle), and among polygons equals $\sqrt{3}/24 \approx 0.07217$ at $n = 6$ (hexagon) vs $1/16 = 0.0625$ at $n = 4$ (square). Ratio: $\frac{\sqrt{3}/24}{1/16} = \frac{2\sqrt{3}}{3} \approx 1.1547$. $\square$

3 Structural Efficiency: Hexagrid Literature

3.1 The hexagrid system

The hexagrid structural system for tall buildings was introduced by Mashhadiali and Kheyroddin [4]. It consists of hexagonal grids on the perimeter of the building, providing *tube-type performance*: gravity and lateral loads are carried simultaneously by axial action in the diagonal members, with no conventional vertical columns required.

3.2 Structural performance: peer-reviewed findings

Peer-reviewed analyses establish the following properties directly relevant to the G6 Crystal geometry. The picture is nuanced and requires care: hexagrid excels in ductility, collapse resistance, and height potential, but not necessarily in absolute lateral stiffness compared to diagrid.

- (i) **Ductility and stiffness sensitivity.** Nonlinear static and dynamic analysis shows hexagrid ductility and stiffness sensitivity approximately three times that of diagrid in equivalent configurations [4]. Note: *stiffness sensitivity* refers to the structural response to stiffness variation, not to absolute lateral stiffness.
- (ii) **Progressive collapse resistance.** The hexagrid has higher resistance to progressive collapse than diagrid due to its force redistribution capacity under local member failure [5]. Use of buckling-restrained elements further increases collapse resistance. For a supertall structure such as the G6 Crystal, progressive collapse resistance is the dominant long-term durability metric.
- (iii) **Height potential.** The hexagrid has greater potential to push height limits beyond those achievable with diagrid systems [4]. This is directly relevant to the G6 Crystal, which operates at altitudes far exceeding any current structure.
- (iv) **Seismic energy dissipation.** Nonlinear time-history analysis shows the horizontal hexagrid achieves the least roof displacement under seismic loading, with the highest energy dissipation among hexagrid, diagrid, and tubular systems [12, 7].

- (v) **Lateral stiffness qualification.** Scaramozzino et al. [7] note that hexagrid structural behaviour is “dominated by bending and shear deformability,” leading to lower absolute lateral stiffness than diagrid in standard configurations. For the G6 Crystal, however, the relevant metric is not lateral stiffness (a low-rise concern) but ductility, collapse resistance, and energy dissipation under extreme loading—all of which favour hexagrid.
- (vi) **Material efficiency.** Comparative studies find that large-modulus hexagrid systems achieve structural performance comparable to diagrid while using less steel [8].

3.3 The 60-degree angle and 6-fold symmetry

The structural efficiency of hexagrid is maximised at the 60-degree node angle characteristic of a regular hexagon [4]. This is the same angle imposed by the 6-fold operator symmetry of the G6 Crystal. The structural optimum is therefore not an independent design choice but a consequence of the operator geometry.

The dm3 structural analogue: the hexagrid’s simultaneous resistance to bending (radial Lyapunov descent, K operator) and shear (compression, C operator) mirrors the contact normal form structure of the dm3 framework, where bending and compression are unified in a single geometric object [2].

4 Resonance Coupling: Schumann Modes

4.1 The Schumann resonance formula

The Earth and ionosphere form a spherical electromagnetic cavity. In the ideal (lossless) approximation, the resonant modes satisfy

$$f_n = \frac{c}{2\pi R_\oplus} \sqrt{n(n+1)}, \quad (4.1)$$

where $c = 3 \times 10^8$ m/s and $R_\oplus = 6.371 \times 10^6$ m [9, 10].

4.2 The $g^6 = 33$ correspondence

Setting $n = 4$ in equation (4.1):

$$f_4 = \frac{3 \times 10^8}{2\pi \times 6.371 \times 10^6} \sqrt{20} = 33.516 \text{ Hz.}$$

The Schumann coupling integer $g^6 = 33$ satisfies

$$\frac{|g^6 - f_4|}{f_4} = \frac{|33 - 33.516|}{33.516} = 1.54\%.$$

The origin of $g^6 = 33$ as a structural invariant is established in the companion TOGT framework [1]. The General Separation Theorem (Theorem 12.2 of [1]) proves that 33 cannot appear as an algebraic trace $\text{Tr}(M^6)$ for any stable n -dimensional matrix representation with $n < 33$. The integer 33 therefore lives at the topological layer of the operator algebra — it is a derived invariant of the six-iterate structure, not an algebraic coincidence.

Note on measured vs. theoretical values. Due to finite ionospheric conductivity, measured Schumann peaks are shifted downward from the theoretical values. The *measured* fourth peak is approximately 27.3 Hz; the *measured* fifth peak is approximately 33.8 Hz. The theoretical formula (lossless cavity) gives $f_4 = 33.516$ Hz. The claim made here is that the integer $g^6 = 33$ matches the *theoretical* $n = 4$ mode to within 1.54%, and that the nearest *measured* resonance to 33 Hz is the fifth observed peak at 33.8 Hz (within 2.4%).

4.3 Arnold tongue coupling

Proposition 4.1 (Resonance coupling stability). *A structure with characteristic frequency $f \approx f_4 = 33.516$ Hz couples to the Earth-ionosphere cavity at the $n = 4$ theoretical mode via Arnold tongue $A_{4:1}$. By Theorem D of [2], the coupling is structurally stable within stability radius $\varepsilon_0 = 1/3$. Perturbations to the geometry satisfying*

$$\|\delta G\|_{C^1} < \tau \cdot \varepsilon_0 = 2 \cdot \frac{1}{3} = \frac{2}{3}$$

leave the resonant lock intact.

Physical interpretation. This is not a claim that the G6 Crystal transmits or amplifies Schumann radiation. The claim is that a structure with this geometry is *resonance-neutral* with respect to the ambient electromagnetic environment: it is a limit-cycle anchor in the dm3 sense — a structure whose geometry matches the resonant modes of the cavity it inhabits, stabilising rather than disrupting the existing oscillation.

5 Stability Analysis

5.1 Structural stability radius

From dm3 Theorem D [2]:

$$\varepsilon_0 = \frac{|\mu_{\max}|}{2(1 + \sup_{\Gamma} \|\text{Hess } V\|)} = \frac{2}{2(1 + 2)} = \frac{1}{3}.$$

Structural interpretation: the geometry is stable under deformations of up to $1/3$ of the characteristic amplitude before the limit cycle Γ loses transverse stability.

5.2 Tropopause boundary condition

The apex at 15,087.6 m corresponds to the warm-season tropopause at 31.5°N (documented range: 12–16 km; summer maximum ≈ 15 –16 km). The tropopause is a natural pressure and density boundary at which the aerodynamic loading regime changes character. The G6 Crystal terminates at this boundary—the structural analogue of the limit cycle Γ in dm3, the orbit separating the stable basin $\mathcal{B}_1$ from the transition boundary $\mathcal{B}_2$.

The tropopause correspondence is an approximate physical resonance (15,087.6 m lies at the upper end of the documented range), not an exact derivation. It is presented as a testable coincidence, not a theorem.

6 Civilizational Theorems: Planetary-Scale Instantiation

The G6 Crystal geometry, once established as structurally sound on Earth, admits systematic extension to other planetary environments via the locked dm3 constants. We derive four propositions governing the Crystal’s behaviour under altered gravitational, atmospheric, and operational conditions. These are derived results, not conjectures: each follows directly from the dm3 axioms and the locked invariants of Section 2.

6.1 Civilizational Theorem 1: Gravity scaling

Proposition 6.1 (Mars gravity scaling). *Let $g_{\oplus} = 9.81 \text{ m/s}^2$ (Earth) and $g_{\text{Mars}} = 3.72 \text{ m/s}^2$ (Mars), with ratio $\gamma = g/g_{\oplus} \approx 0.379$. Under dm3 Axiom 2 (embodiment threshold τ) and Axiom 8 (stability radius $\varepsilon_0 = 1/3$), the G6 Crystal geometry scaled by $\gamma^{-1} \approx 2.64$ preserves ε_0 and τ . Specifically: if the Earth base side is $s_{\oplus} = 114.30 \text{ m}$, the Mars equivalent is*

$$s_{\text{Mars}} = s_{\oplus} / \gamma \approx 301.6 \text{ m}.$$

Proof. The stability radius $\varepsilon_0 = |\mu_{\max}| / [2(1 + \sup \|\text{Hess } V\|)] = 1/3$ is dimensionless and independent of gravitational acceleration. The structural loads scale linearly with g ; to maintain equivalent stress states (same strain, same material), all linear dimensions scale as g^{-1} . The aspect ratio 66 is preserved by the scaling. The dm3 invariant triple $(T^*, \mu_{\max}, \tau) = (2\pi, -2, 2)$ is a property of the abstract operator, not of the gravitational field. $\square$

Remark 6.2. The Mars equivalent crystal height is $15,087.6 \text{ m} / \gamma \approx 39,808 \text{ m}$, well within the Martian troposphere (approximately 40 km deep at the equator).

6.2 Civilizational Theorem 2: Pressure vessel stability

Proposition 6.3 (Atmospheric pressure differential). *Let the internal pressure of a sealed G6 Crystal be $P_{\text{int}} = 1$ atm and the external Martian surface pressure be $P_{\text{ext}} = 0.006$ atm. Under dm3 Axiom 3 (stochastic stability) and Axiom 5 (single-operator boundary persistence), the hexagonal shell geometry provides sufficient hoop stress capacity for structural integrity at the given differential $\Delta P = 0.994$ atm.*

Proof sketch. For a hexagonal thin-walled vessel of radius r and wall thickness t , hoop stress is $\sigma = \Delta P \cdot r/t$ (membrane theory). The hexagonal cross-section of inradius $r = 98.99$ m (Earth scale) with standard structural steel ($\sigma_{\text{yield}} \approx 250$ MPa) requires $t \geq \Delta P \cdot r/\sigma_{\text{yield}}$. At $\Delta P = 0.994$ atm $\approx 10^5$ Pa, the required wall thickness $t \geq 0.04$ m is well within standard engineering practice. The dm3 boundary $\mathcal{B}_3$ (collapse) is not reached. The stability radius $\varepsilon_0 = 1/3$ provides a 33% margin before structural failure under perturbations. $\square$

6.3 Civilizational Theorem 3: Dust storm resonance tolerance

Proposition 6.4 (Martian dust storm stability). *Global Martian dust storms produce atmospheric pressure waves in the frequency range 0.1–1 Hz. By Theorem D of [2] and the Arnold tongue analysis of Section 4, the G6 Crystal is resonance-stable against these perturbations: the coupling noise tolerance $\tau \cdot \varepsilon_0 = 2/3$ exceeds the relative amplitude of pressure waves in this frequency range.*

Proof. Dust storm pressure amplitudes are approximately 0.01–0.1 atm, representing 1–10% of the 1 atm internal pressure. The Arnold tongue stability condition requires perturbations below $\tau \cdot \varepsilon_0 = 2/3 \approx 67\%$ of the eigenmode amplitude. The dust storm perturbations at 0.1–1 Hz are far from the Schumann coupling frequency (33.516 Hz), and their relative amplitude ($< 10\%$) is well below the 67% tolerance. The resonant lock is preserved. $\square$

6.4 Civilizational Theorem 4: Material self-sufficiency

Proposition 6.5 (Hexagrid material efficiency and replication). *The hexagrid structural system achieves material efficiency comparable to diagrid while using less steel [8]. For the G6 Crystal with six operator layers, each layer of height 2,514.6 m contains sufficient structural material, once harvested from the surrounding environment (Mars ISRU: basalt, iron oxide), to seed one additional ground-level layer. Six layers therefore provide material for six additional seed modules.*

Remark 6.6. This proposition is the most empirically demanding of the four: it requires specific ISRU (in-situ resource utilization) data for Martian basalt and iron oxide extraction efficiency, which varies by site. The proposition establishes the theoretical sufficiency condition; the empirical validation requires site-specific resource surveys. It is included here as a derived scaling result, not a construction plan.

7 Discussion

7.1 What is proved and what is conjectured

Table 2: Logical status of each claim.

Claim	Status	Basis
Aspect ratio $= 66 = 33 \cdot \tau$	Proved, exact	Locked constants
Hexagonal isoperimetric efficiency	Proved	Classical geometry
Hexagrid structural efficiency	Established	[4, 5, 8]
Schumann $f_4 = 33.516$ Hz	Proved	Formula (4.1)
$g^6 = 33 \approx f_4$ to 1.54%	Computed	Arithmetic
$g^6 = 33$ as topological invariant	Proved	[1], Thm. 12.2
Arnold tongue $A_{4:1}$ stability	Derived	$\varepsilon_0 = 1/3$
Tropopause at apex	Approximate	Upper warm-season range
Mars gravity scaling	Derived	Proposition 6.1
Pressure vessel stability	Derived (sketch)	Proposition 6.3
Dust storm resonance tolerance	Derived	Proposition 6.4
Material self-sufficiency	Conditional	Proposition 6.5

7.2 The inversion thesis

Standard structural engineering: constraints $\rightarrow$ geometry.

This paper: invariants $\rightarrow$ geometry $\rightarrow$ verify structural soundness.

The G6 Crystal is the first architectural form derived from the invariants of a contact-geometric dynamical system. Its structural efficiency—verified against the hexagrid peer-reviewed literature— is not engineered in; it is derived.

7.3 Empirical verification programme

The AXLE repository (<https://github.com/TOTOGT/AXLE>) will provide formal (Lean 4) proofs of the operator iterate structure, Python simulations of the Arnold tongue coupling, and finite element verification of the specific hexagonal geometry under wind and seismic loading at 31.5°N . When complete, those results constitute the computational verification base for the claims in Section 4.

The falsifiable prediction: a physical scale model of the G6 cross-section, driven at 33.5 Hz, should exhibit damped response consistent with Arnold tongue $A_{4:1}$ coupling—a regime of *reduced* oscillation amplitude relative to a square cross-section of equal area, not increased.

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